

KAVIENAN JEGATHEESAN

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SUMMARY

Final-year Computer Science and Engineering undergraduate at the University of Moratuwa, with a strong passion for Cybersecurity, AI and Cryptography. Currently pursuing research on Trustworthy & Reliable AI, Emerging Threats of Agentic Systems, and Software Supply Chain Security.

EDUCATION

- **University of Moratuwa** Mar 2023 – Present
Bachelor of Science (Honours) in Engineering Moratuwa, Sri Lanka
 - Specialization: Computer Science and Engineering (Cybersecurity Stream)
 - GPA: 3.68/4.0 [[View Transcript](#)]
- **Royal College** Jan 2008 – Mar 2022
Primary and Secondary Education Colombo, Sri Lanka
 - G.C.E. Advanced Level: 3 A's in the Physical Science Stream (Z-score: 2.1595)

EXPERIENCE

- **Teaching Assistant** Aug 2026 – Present
CS2033, Data Communication and Networking Lab, University of Moratuwa Part-time, On-site
 - Teach undergraduates core networking concepts, including IP addressing, TCP/IP, and the OSI layer stack, and conduct hands-on lab practicals using real Cisco switches and routers.
- **Intern Cybersecurity Engineer** Nov 2025 – May 2026
WSO2 Lanka PVT Limited [🔗](#) Full-time, On-site
 - Developed a scheduled **scanning system that evaluates GitHub repositories** against security, maintenance, and community standards, with a monitoring dashboard.
 - Built an **AI-powered service to refine security advisory content** to public announcement standards, along with a companion Chrome Extension to consume it.
- **Full-Stack Developer** June 2024 – May 2025
Simology Limited, United Kingdom [🔗](#) Part-Time, Remote
 - Led a team of developers in building end-to-end **eSIM sales platform website backends and mobile apps**.

RESEARCH CONTRIBUTIONS

- **"When Tools Lie: Reliability of Mathematical Agents Under Corrupted Tool Feedback"** 2026
Accepted (Poster), 6th Workshop on Mathematical Reasoning and AI, NeurIPS 2026 First Author
 - Built a controlled corruption framework with a hidden interceptor that silently replaces tool outputs with plausible but incorrect results, and evaluated agents on 31 problems under four verification designs.
 - Showed that corruption drops accuracy from 100% to 72.4%, mandatory reflection restores it to 100%, and optional verification helps only when models choose to invoke it, so checking frequency drives robustness.
- **Random Sampling for Linear Learned Index Fitting** 2026
Submitted, ERU Research Symposium 2026, University of Moratuwa First Author
 - Proposed fitting a linear learned index on a small random subset of keys, making fitting cost independent of dataset size and resistant to adversarial outlier poisoning, while preserving prediction accuracy and speed.
 - Benchmarked six distributions and three sizes, matching exact-fit accuracy and speed on clean data and achieving up to 319x lower error and 2.7x faster queries under poisoning, while deterministic sampling offered no protection.
- **"When Revocation Is Not Enough: Identity Scope and State Freshness in ERC-8004"** 2026
Submitted, 18th Australasian Information Security Conference (AISC 2027) First Author
 - Models an append-only RSA accumulator as revocation registry for ERC-8004 agent identities, testing non-membership proofs across prover-supplied, cached, and current states to isolate freshness failures.
 - Finds a revoked agent can re-register under a fresh identity and pass verification, and that cached-state verifiers keep accepting revoked agents until refresh, then proposes a bounded-freshness policy to cap this exposure.
- **Deriving Execution Environments for Dynamic Analysis of Packages** 2026
Manuscript in preparation, target venue 36th USENIX Security Symposium Co-Author
 - Parses package source into a syntax tree to extract the machine conditions malicious packages gate their payloads on, then derives a covering-array based set of execution environments guaranteed to combine those conditions.
 - Provisions each environment for an existing dynamic scanner and compares detections against a single reference environment and a random baseline of equal size to test whether the derived set actually helps.

PROJECTS

- **Code Scanner AI - Multi-Agent AI Security Analysis Tool** [🔗] 2026
Technologies: Next.js, TypeScript, React, Tailwind CSS, OpenAI API, Anthropic API, Docker Individual Project
 - Built a three-agent pipeline that discovers API endpoints, generates OWASP-mapped security checklists, and inspects code for vulnerabilities like SQL injection, XSS, and command injection.
 - Automatic framework detection across Next.js, Django, Flask, FastAPI, and Spring Boot, with a security scoring dashboard and real-time analysis logs.
- **TrueVote - Blockchain-Based Decentralized Voting Platform** [🔗] 2025
Technologies: Web3.js, Ganache, Solidity, React.js, Node.js, MySQL, AWS S3 Group Project
 - Ethereum smart contracts using Solidity, enabling decentralized and immutable campaign data storage.
 - Solidity event emitters and Web3.js listeners to stream real-time vote tallies to the frontend with minimal latency.
- **SafeCom - AI-Powered Spam & Scam Detection** [🔗] 2025
Technologies: Flutter, Kotlin, OpenAI APIs, Next.js, Node.js Group Project
 - Analyzes call duration, call history, contact details, and message content to identify suspicious activity.
 - Cross-references caller identities with open-source intelligence to verify if the person is who they claim to be.

SKILLS

- **Programming Languages:** Python, TypeScript, SQL, Java, C, C++, Solidity, Bash
- **AI & LLMs:** OpenAI API, Anthropic API, LangChain/LangGraph, Prompt Engineering, MCP
- **Cybersecurity Tools:** OWASP ZAP, Wireshark, Nmap, Burp Suite
- **Frameworks & Cloud:** React.js, Next.js, Node.js, Docker, AWS, Google Cloud

COMPETITIONS AND AWARDS

- **Winners of DreamShot AI Hackathon** Jan 2025
3-day AI Hackathon, organized by DreamSpace Academy Sri Lanka
 - Awarded the **Best AI Solution Award** for the project **SafeCom - AI Scam Detection App**.
- **Winners of Idealize 4.0** Aug 2024
Idealize 4.0 Open Category, organized by AIESEC in the University of Moratuwa
 - Awarded **Winner** and received the **Idealizer Award & Trendsetter Title** for the project **Localize Sri Lanka**.
- **President's Scout Award** Feb 2022
Scouting, National Level

VOLUNTEER EXPERIENCE

- **Volunteer - 1st ADScAI Symposium 2025** Mar 2025 – May 2025
CSE40 - 1st Applied Data Science & Artificial Intelligence Symposium 2025
 - As part of the Publications Committee, composed the proceedings document from authors' camera-ready papers.
- **Volunteer - Cyber Rush 2025** Mar 2025 – May 2025
CSE40 - CodeJam 2025 Capture the Flag Competition
 - Collaborated with the development team to design original cryptography and web vulnerability challenges.

LEADERSHIP EXPERIENCE

- **Student Representative** Jan 2025 – June 2026
Dept. of Computer Science and Engineering, University of Moratuwa
- **Webmaster of CSESS** July 2025 – May 2026
Computer Science and Engineering Students Society, University of Moratuwa
- **Senior Prefect** Mar 2022 – Mar 2023
Royal College

ADDITIONAL INFORMATION

Languages: English (Professional working proficiency), Tamil (Native proficiency), Sinhala (Professional working proficiency)

REFERENCES

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